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DROPLET DISCHARGING APPARATUS, DROPLET DISCHARGING METHOD AND MANUFACTURING METHOD FOR DISPLAY DEVICE

Abstract

A droplet discharging apparatus may include a transfer part moving a substrate including a first target position in a first direction, an inkjet head disposed on the transfer part and discharging a first droplet to the first target position of the substrate, a first camera positioned in a second direction opposite to the first direction of the inkjet head, the first camera acquiring a first image corresponding to the first target position of the substrate, a second camera positioned in the first direction of the inkjet head, the second camera acquiring a second image corresponding to the first target position of the substrate, and an image processor receiving the first image and the second image, and comparing the first image and the second image to inspect a discharge state of the first droplet.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This is a divisional application of U.S. patent application Ser. No. 17/826,831, filed May 27, 2022 (now pending) the disclosure of which is incorporated herein by reference in its entirety. U.S. patent application Ser. No. 17/826,831 claims priority to and the benefits of Korean Patent Application No. 10-2021-0120436 under 35 U.S.C. § 119, filed Sep. 9, 2021, in the Korean Intellectual Property Office, the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Technical Field

[0002] Embodiments of the disclosure relate to a droplet discharging apparatus used in a printing process, a droplet discharging method used in a printing process, and a manufacturing method for a display device.

2. Description of the Related Art

[0003] In manufacturing a display device, an emitting layer, a monomer layer of a thin film encapsulation structure, a color filter layer, and the like may be formed through a printing process.

[0004] The printing process may be performed in such a way that an inkjet head disposed on a two-dimensional plane spaced apart from a substrate by a distance discharges a droplet onto the substrate. The substrate may have an opening capable of accommodating the droplet. The droplet discharged from the inkjet head may impact the opening.

[0005] In the related art, a state of the droplet may be inspected before performing the printing process, but there is a problem in that it is not possible to check whether the droplet impacts during the printing process.

[0006] It is to be understood that this background of the technology section is, in part, intended to provide useful background for understanding the technology. However, this background of the technology section may also include ideas, concepts, or recognitions that were not part of what was known or appreciated by those skilled in the pertinent art prior to a corresponding effective filing date of the subject matter disclosed herein.

SUMMARY

[0007] Embodiments of the disclosure provide a droplet discharging apparatus capable of checking whether the droplet impacts while the printing process may be performed.

[0008] Embodiments of the disclosure also provide a droplet discharging method capable of checking whether the droplet impacts while the printing process may be performed.

[0009] Embodiments of the disclosure also provide a manufacturing method for a display device including the droplet discharging method.

[0010] Additional features of the disclosure will be set forth in the description which follows, and

in part will be apparent from the description, or may be learned by practice of the disclosure.

[0011] A droplet discharging apparatus according to an embodiment may include a transfer part moving a substrate including a first target position in a first direction, an inkjet head disposed on the transfer part and discharging a first droplet to the first target position of the substrate, a first camera positioned in a second direction opposite to the first direction of the inkjet head, the first camera acquiring a first image corresponding to the first target position of the substrate, a second camera positioned in the first direction of the inkjet head, the second camera acquiring a second image corresponding to the first target position of the substrate, and an image processor receiving the first image and the second image, and comparing the first image and the second image to inspect a discharge state of the first droplet.

[0012] In an embodiment, the first camera may acquire the first image at a first timing, the inkjet head may discharge the first droplet at a second timing later than the first timing, and the second camera may acquire the second image at a third timing later than the second timing.

[0013] In an embodiment, at the first timing, the second timing, and the third timing, the transfer part may be moving the substrate in the first direction.

[0014] In an embodiment, the first image may be an image corresponding to the first target position before the first droplet may be discharged, and the second image may be an image corresponding to the first target position after the first droplet may be discharged.

[0015] In an embodiment, the substrate may further include a second target position spaced apart from the first target position in the second direction, the first camera may acquire a third image corresponding to the second target position of the substrate, the inkjet head may discharge a second droplet to the second target position of the substrate, the second camera may acquire a fourth image corresponding to the second target position of the substrate, and the image processor may receive the third image and the fourth image, and may compare the third image and the fourth image to inspect a discharge state of the second droplet.

[0016] In an embodiment, the first camera may acquire the third image at a fourth timing later than the first timing, the inkjet head may discharge the second droplet at a fifth timing later than the fourth timing, and the second camera may acquire the fourth image at a sixth timing later than the fifth timing.

[0017] In an embodiment, the first camera, the inkjet head, and the second camera may be physically connected to each other and fixed.

[0018] In an embodiment, each of the first camera and the second camera may be a line scan camera that photographs in units of lines.

[0019] In an embodiment, each of the first camera and the second camera may be an area scan camera that photographs in units of areas.

[0020] In an embodiment, the droplet discharging apparatus may further include a controller communicating with the image processor and stopping a printing process in case that it is determined that the first droplet is not seated at the first target position.

[0021] In an embodiment, a direction in which the first camera and the second camera photograph may be a third direction perpendicular to the substrate.

[0022] In an embodiment, the droplet discharging apparatus may further include an illumination including a light source irradiating light in a direction intersecting the third direction, and a half mirror reflecting the light in the third direction.

[0023] In an embodiment, the second camera may receive a first reflected light reflected by the substrate, the second camera receives a second reflected light reflected by the first droplet, the first reflected light may have a first intensity, and the second reflected light may have a second intensity greater than the first intensity.

[0024] A droplet discharging method according to an embodiment may include moving a substrate including a first target position in a first direction, acquiring a first image corresponding to the first target position of the substrate at a first timing, discharging a first droplet to the first target position

of the substrate at a second timing, acquiring a second image corresponding to the first target position of the substrate at a third timing, and comparing the first image and the second image to inspect a discharge state of the first droplet.

[0025] In an embodiment, at the first timing, the second timing, and the third timing, the substrate may be moved in the first direction.

[0026] In an embodiment, the first image may be an image corresponding to the first target position before the first droplet may be discharged, and the second image may be an image corresponding to the first target position after the first droplet may be discharged.

[0027] In an embodiment, the droplet discharging method may further include stopping a printing process in case that it is determined that the first droplet is not seated at the first target position.

[0028] In an embodiment, the droplet discharging method may further include acquiring a third image corresponding to a second target position of the substrate, the second target position spaced apart from the first target position in a second direction opposite to the first direction, discharging a second droplet to the second target position of the substrate, acquiring a fourth image corresponding to the second target position of the substrate, and comparing the third image and the fourth image to inspect a discharge state of the second droplet.

[0029] A manufacturing method for a display device according to an embodiment may include forming an anode electrode on a base substrate, forming a pixel defining layer on the base substrate, the pixel defining layer having an opening exposing the anode electrode, moving the base substrate in a first direction, acquiring a first image corresponding to the opening of the base substrate, forming an emission layer by discharging a droplet including an organic light emitting material into the opening of the base substrate, acquiring a second image corresponding to the opening of the base substrate, and comparing the first image and the second image to inspect a discharge state of the droplet.

[0030] In an embodiment, the first image may be an image corresponding to the opening before the droplet may be discharged, and the second image may be an image corresponding to the opening after the droplet may be discharged.

[0031] According to the droplet discharging apparatus, the droplet discharging method, and the manufacturing method of the display device according to embodiments of the disclosure, in the time period during which the printing process may be performed, an image corresponding to the target position may be acquired immediately before the droplet may be discharged to the target position, another image corresponding to the target position may be acquired immediately after the droplet may be discharged to the target position, and the images may be compared. Accordingly, it may be possible to check in real time whether the droplet impacts while the printing process may be performed.

[0032] It is to be understood that both the foregoing general description and the following detailed description are examples and explanatory and are intended to provide further explanation of the disclosure as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0033] The accompanying drawings, which are included to provide a further understanding of the disclosure and are incorporated in and constitute a part of this specification, illustrate embodiments of the disclosure, and together with the description serve to explain the disclosure.

[0034] FIG. 1 is a schematic view illustrating a droplet discharging apparatus according to an embodiment.

[0035] FIG. 2 is a schematic perspective view illustrating an example of a transfer part and an inkjet head included in the droplet discharging apparatus of FIG. 1.

[0036] FIG. 3 is a schematic plan view illustrating an example of the transfer part of FIG. 2.

[0037] FIG. 4 is a schematic view illustrating an example of an illumination included in the droplet discharging apparatus of FIG. 1.

[0038] FIGS. 5 to 7 are schematic views for explaining an operation of the droplet discharging apparatus of FIG. 1.

[0039] FIG. 8 is a schematic view illustrating a first image acquired by a first camera of FIG. 5.

[0040] FIG. 9 is a schematic view illustrating a second image acquired by a second camera of FIG. 7.

[0041] FIG. 10 is a schematic view illustrating an example of a first full image including the first image of FIG. 8.

[0042] FIG. 11 is a schematic view illustrating an example of a second full image including the second image of FIG. 9.

[0043] FIGS. 12 to 14 are schematic views for explaining an additional operation of the droplet discharging apparatus of FIG. 1.

[0044] FIG. 15 is a schematic view illustrating a third image acquired by a first camera of FIG. 12.

[0045] FIG. 16 is a schematic view illustrating a fourth image acquired by a second camera of FIG. 14.

[0046] FIG. 17 is a schematic view illustrating an example of a first full image further including a third image of FIG. 15.

[0047] FIG. 18 is a schematic view illustrating an example of a second full image further including a fourth image of FIG. 16.

[0048] FIGS. 19 to 22 are schematic views illustrating a manufacturing method for a display device according to an embodiment.

DETAILED DESCRIPTION OF THE EMBODIMENTS

[0049] The disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which embodiments are shown. This disclosure may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art.

[0050] In the drawings, sizes, thicknesses, ratios, and dimensions of the elements may be exaggerated for ease of description and for clarity. Like numbers refer to like elements throughout.

[0051] As used herein, the singular forms, “a,” “an,” and “the” are intended to include the plural forms as well, unless the context clearly indicates otherwise.

[0052] The terms “comprises,” “comprising,” “includes,” and/or “including,” “has,” “have,” and/or “having,” and variations thereof when used in this specification, specify the presence of stated features, integers, steps, operations, elements, components, and/or groups thereof, but do not preclude the presence or addition of one or more other features, integers, steps, operations, elements, components, and/or groups thereof.

[0053] In the specification and the claims, the term “and/or” is intended to include any combination of the terms “and” and “or” for the purpose of its meaning and interpretation. For example, “A and/or B” may be understood to mean “A, B, or A and B.” The terms “and” and “or” may be used in the conjunctive or disjunctive sense and may be understood to be equivalent to “and/or.”

[0054] It will be understood that, although the terms first, second, etc., may be used herein to describe various elements, these elements should not be limited by these terms. These terms are only used to distinguish one element from another element. For example, a first element may be referred to as a second element, and similarly, a second element may be referred to as a first element without departing from the scope of the disclosure.

[0055] It will be understood that when an element (or a region, a layer, a portion, or the like) is referred to as “being on,” “connected to” or “coupled to” another element in the specification, it can be directly disposed on, connected or coupled to another element mentioned above, or

intervening elements may be disposed therebetween. It will be understood that the terms “connected to” or “coupled to” may include a physical or electrical connection or coupling. [0056] Unless otherwise defined, all terms (including technical and scientific terms) used herein have the same meaning as commonly understood by one of ordinary skill in the art to which the disclosure pertains. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and will not be interpreted in an idealized or overly formal sense unless expressly so defined herein.

[0057] FIG. 1 is a schematic view illustrating a droplet discharging apparatus according to an embodiment. FIG. 2 is a schematic perspective view illustrating an example of a transfer part and an inkjet head included in the droplet discharging apparatus of FIG. 1. FIG. 3 is a schematic plan view illustrating an example of the transfer part of FIG. 2. FIG. 4 is a schematic view illustrating an example of an illumination included in the droplet discharging apparatus of FIG. 1.

[0058] Referring to FIGS. 1 to 4, a droplet discharging apparatus **1000** according to an embodiment may include a transfer part **100**, an inkjet head **200**, a first camera **300**, a second camera **400**, an image processor **500**, a controller **600**, and an illumination **700**.

[0059] The transfer part **100** may transfer a substrate **50** in a first direction **D1**. Although it has been described that the transfer part **100** can transfer the substrate **50** in the first direction **D1**, embodiments are not limited thereto. For example, the transfer part **100** may transfer the substrate **50** in a direction different from the first direction **D1**. For example, the transfer part **100** may reciprocally transfer the substrate **50** in the first direction **D1** and in a second direction **D2** opposite to the first direction **D1**.

[0060] The inkjet head **200** may be disposed on the transfer part **100**. The inkjet head **200** may discharge a droplet to the substrate **50** transferred by the transfer part **100**.

[0061] The substrate **50** may include a first area **51a**. The first area **51a** may be a target area from which the droplet may be discharged while the transfer part **100** moves the substrate **50** in the first direction **D1**.

[0062] The substrate **50** may further include a second area **51b** different from the first area **51a**. The second area **51b** may be a target area from which the droplet may be discharged while the transfer part **100** moves the substrate **50** in the second direction **D2**.

[0063] FIG. 2 illustrates that the first area **51a** and the second area **51b** constitute an upper surface **51** of the substrate **50**, but embodiments are not limited thereto. For example, the upper surface **51** may be constituted with more areas.

[0064] Hereinafter, for convenience of understanding, it may be assumed that the transfer part **100** moves the substrate **50** in the first direction **D1**. Accordingly, a process in which the inkjet head **200** discharges the droplet to the second area **51b** of the substrate **50** will not be described, but those skilled in the art will be able to easily understand the process.

[0065] In case that the inkjet head **200** discharges the droplet, the substrate **50** may receive the droplet. The inkjet head **200** may discharge the droplet multiple times while the transfer part **100** moves the substrate **50** in the first direction **D1**.

[0066] The droplet may include a first droplet **10**. The first droplet **10** may be a droplet discharged by the inkjet head **200** at any of the multiple times. The first droplet **10** may be plural.

[0067] The substrate **50** may include a first target position **60**. The first target position **60** may be positioned in the first area **51a** of the substrate **50**. The first target position **60** may be a position where the substrate **50** can receive the first droplet **10**. If the first droplet **10** may be plural, the first target position **60** may also be plural. The first target position **60** may mean a position that may be parallel to each other in a direction crossing the first direction **D1**.

[0068] The droplet may further include a second droplet (e.g., a second droplet **20** of FIG. 13). The second droplet may be a droplet discharged by the inkjet head **200** at another of the multiple times. The second droplet may be plural.

[0069] The substrate **50** may further include a second target position **70** spaced apart from the first target position **60** in the second direction **D2**. The second target position **70** may be positioned in the first area **51a** of the substrate **50**. The second target position **70** may be a position where the substrate **50** can receive the second droplet. If the second droplet is plural, the second target position **70** may also be plural. The second target position **70** may mean another position that may be parallel to each other in a direction crossing the first direction **D1**.

[0070] The first camera **300** may be positioned in the second direction **D2** of the inkjet head **200**. The first camera **300** may acquire a first image (e.g., a first image **301a** of FIG. **8**) corresponding to the first target position **60** of the substrate **50**. The first image may be an image corresponding to the first target position **60** before the first droplet **10** may be discharged.

[0071] The second camera **400** may be positioned in the first direction **D1** of the inkjet head **200**. The second camera **400** may acquire a second image (e.g., a second image **401a** of FIG. **9**) corresponding to the first target position **60** of the substrate **50**. The second image may be an image corresponding to the first target position **60** after the first droplet **10** may be discharged. The first image and the second image will be described in detail later.

[0072] In an embodiment, the inkjet head **200**, the first camera **300**, and the second camera **400** may be connected to each other and fixed. For example, the first camera **300**, the inkjet head **200**, and the second camera **400** may be supported by a support (not illustrated).

[0073] In another embodiment, the inkjet head **200**, the first camera **300**, and the second camera **400** may be fixed while being separated from each other. For example, the inkjet head **200** may be supported by a support (not illustrated), the first camera **300** may be supported by another support (not illustrated), and the second camera **400** may be supported by another support (not illustrated).

[0074] A direction in which the first camera **300** and the second camera **400** photograph may be a third direction **D3** perpendicular to the substrate **50**.

[0075] In an embodiment, each of the first camera **300** and the second camera **400** may be a line scan camera that photographs in units of lines. Each of the first image and the second image may be photographed in units of lines.

[0076] In another embodiment, each of the first camera **300** and the second camera **400** may be an area scan camera that photographs in units of areas. Each of the first image and the second image may be photographed in units of areas.

[0077] The image processor **500** may inspect a discharge state of the droplet discharged from the inkjet head **200**. The image processor **500** may receive the first image and the second image, and may compare the first image and the second image to inspect the discharge state (e.g., impact or not) of the first droplet **10**.

[0078] The controller **600** may communicate with the image processor **500**. The controller **600** may control the transfer part **100** and the inkjet head **200**. A printing process may be performed by controlling the transfer part **100** and the inkjet head **200**.

[0079] The illumination **700** may irradiate light to the substrate **50**. In an embodiment, as illustrated in FIG. **4**, the illumination **700** may include a first illumination **700a** connected to the first camera **300** and a second illumination **700b** connected to the second camera **400**.

[0080] The illumination **700** may include a light source **710** and a half mirror **720**. The light source **710** may include a first light source **710a** installed in the first illumination **700a** and a second light source **710b** installed in the second illumination **700b**. The half mirror **720** may include a first half mirror **720a** installed in the first illumination **700a** and a second half mirror **720b** installed in the second illumination **700b**.

[0081] The first light source **710a** may irradiate light in a direction crossing the third direction **D3**. The first half mirror **720a** may reflect the light in the third direction **D3**. The light may be reflected by the substrate **50** to reach the first camera **300**. The first camera **300** may receive a first reflected light reflected by the substrate **50**. The first camera **300** may include a first image sensor. The first image sensor may sense the first reflected light.

[0082] Similarly, the second light source **710b** may irradiate light in a direction crossing the third direction **D3**. The second half mirror **720b** may reflect the light in the third direction **D3**. The light may be reflected by the substrate **50** or the droplet to reach the second camera **400**. The second camera **400** may receive a second reflected light reflected by the substrate **50** or the droplet. The second camera **400** may include a second image sensor. The second image sensor may sense the second reflected light.

[0083] The second reflected light may be divided into a second-first reflected light reflected by the substrate **50** and a second-second reflected light reflected by the droplet.

[0084] The first camera **300** may receive the first reflected light reflected by the substrate **50**. The second camera **400** may receive the second-first reflected light reflected by the substrate **50** and the second-second reflected light reflected by the droplet. The second-second reflected light may be light reflected by the first droplet **10** or the second droplet.

[0085] Each of the first reflected light and the second-first reflected light reflected by the substrate **50** may have a first intensity. The first intensity may vary according to a wavelength of the light irradiated from the light source **710**.

[0086] The second-second reflected light reflected by the droplet may have a second intensity. The second intensity may vary according to the wavelength of the light irradiated from the light source **710**.

[0087] The light source **710** may have a target wavelength range. In case that the light having the target wavelength range is irradiated from the light source **710**, the first intensity may be smaller than the second intensity. For example, the target wavelength range may be a wavelength range in which a difference between the first intensity and the second intensity may be relatively large.

[0088] As will be described later in detail with reference to FIG. **9**, the second image (e.g., the second image **401a** of FIG. **9**) may include a first portion in which the first droplet **10** appears and a second portion in which the substrate **50** appears. In case that the light having the target wavelength range is irradiated from the light source **710**, the first portion in which the first droplet **10** appears may appear brighter than the second portion in which the substrate **50** appears. Accordingly, in the second image, the first portion and the second portion may be clearly distinguished.

[0089] FIGS. **5** to **7** are schematic views for explaining an operation of the droplet discharging apparatus of FIG. **1**. FIG. **8** is a schematic view illustrating a first image acquired by the first camera of FIG. **5**. FIG. **9** is a schematic view illustrating a second image acquired by the second camera of FIG. **7**.

[0090] Hereinafter, with reference to FIGS. **5** to **9**, a droplet discharging method using the droplet discharging apparatus **1000** according to an embodiment will be described.

[0091] Referring to FIGS. **5** to **7**, the transfer part **100** may transfer the substrate **50** in the first direction **D1**. Accordingly, the first camera **300** and the second camera **400** may photograph the first target position **60** of the substrate **50** at different timings. For example, the first camera **300** may photograph the first target position **60** of the substrate **50** earlier than the second camera **400**.

[0092] Specifically, as illustrated in FIG. **5**, the first camera **300** may acquire the first image **301a** corresponding to the first target position **60** at a first timing. At the first timing, the substrate **50** may be moving in the first direction **D1** by the transfer part **100**.

[0093] As illustrated in FIGS. **5** and **8**, the first image **301a** may be an image corresponding to the first target position **60** before the first droplet **10** may be discharged. In other words, a scene in which the first droplet **10** does not impact the first target position **60** of the substrate **50** may appear in the first image **301a**.

[0094] As illustrated in FIG. **6**, the inkjet head **200** may discharge the first droplet **10** at a second timing later than the first timing. The first droplet **10** may impact the first target position **60** of the substrate **50**. At the second timing, the substrate **50** may be moving in the first direction **D1** by the transfer part **100**.

[0095] As illustrated in FIG. 7, the second camera **400** may acquire the second image **401a** corresponding to the first target position **60** at a third timing later than the second timing. At the third timing, the substrate **50** may be moving in the first direction **D1** by the transfer part **100**.

[0096] As illustrated in FIGS. 7 and 9, the second image **401a** may be an image corresponding to the first target position **60** after the first droplet **10** may be discharged. In other words, a scene in which the first droplet **10** impacts the first target position **60** of the substrate **50** may appear in the second image **401a**.

[0097] The image processor **500** may receive the first image **301a** and the second image **401a**, and compare the first image **301a** and the second image **401a** to inspect a discharge state (e.g., impact or not) of the first droplet **10**. For example, the image processor **500** may compare the first image **301a** and the second image **401a** to check whether the first droplet **10** may be properly seated at the first target position **60**. In case that it is determined that the first droplet **10** is not seated at the first target position **60**, the controller **600** may stop the printing process.

[0098] In an embodiment, as illustrated in FIG. 1, there may be an encoder **40** around the transfer part **100**. The encoder **40** may detect an operation of the transfer part **100**. The encoder **40** may be used to coordinate a moving speed of the transfer part **100** and a shooting speed of the first camera **300** and the second camera **400**.

[0099] FIG. 10 is a schematic view illustrating an example of a first full image including the first image of FIG. 8. FIG. 11 is a schematic view illustrating an example of a second full image including the second image of FIG. 9.

[0100] Referring to FIGS. 2, 8 and 10, the first camera **300** may acquire a first full image **301** corresponding to the first area **51a** of the substrate **50** by photographing the substrate **50**.

[0101] Since the first target position **60** may be positioned in the first area **51a** of the substrate **50**, the first full image **301** acquired by the first camera **300** may include the first image **301a**.

[0102] Referring to FIGS. 2, 9 and 11, the second camera **400** may acquire a second full image **401** corresponding to the first area **51a** of the substrate **50** by photographing the substrate **50**.

[0103] Since the first target position **60** may be positioned in the first area **51a** of the substrate **50**, the second full image **401** acquired by the second camera **400** may include the second image **401a**.

[0104] FIGS. 12 to 14 are schematic views for explaining an additional operation of the droplet discharging apparatus of FIG. 1. FIG. 15 is a schematic view illustrating a third image acquired by a first camera of FIG. 12. FIG. 16 is a schematic view illustrating a fourth image acquired by a second camera of FIG. 14.

[0105] Referring to FIG. 12, the first camera **300** may acquire the third image **301b** corresponding to the second target position **70** at a fourth timing later than the first timing. At the fourth timing, the substrate **50** may be moving in the first direction **D1** by the transfer part **100**.

[0106] As illustrated in FIG. 15, the third image **301b** may be an image corresponding to the second target position **70** before the second droplet **20** may be discharged. In other words, a scene in which the second droplet **20** does not impact the second target position **70** of the substrate **50** may appear in the third image **301b**.

[0107] Referring to FIG. 13, the inkjet head **200** may discharge the second droplet **20** at a fifth timing later than the fourth timing. The second droplet **20** may impact the second target position **70** of the substrate **50**. At the fifth timing, the substrate **50** may be moving in the first direction **D1** by the transfer part **100**.

[0108] Referring to FIG. 14, the second camera **400** may acquire the fourth image **401b** corresponding to the second target position **70** at a sixth timing later than the fifth timing. At the sixth timing, the substrate **50** may be moving in the first direction **D1** by the transfer part **100**.

[0109] As illustrated in FIG. 16, the fourth image **401b** may be an image corresponding to the second target position **70** after the second droplet **20** may be discharged. In other words, a scene in which the second droplet **20** impacts the second target position **70** of the substrate **50** may appear in the fourth image **401b**.

[0110] The image processor **500** may receive the third image **301b** and the fourth image **401b**, and compare the third image **301b** and the fourth image **401b** to inspect a discharge state (e.g., impact or not) of the second droplet **20**. For example, the image processor **500** may compare the third image **301b** and the fourth image **401b** to check whether the second droplet **20** may be properly seated at the second target position **70**. In case that it is determined that the second droplet **20** is not seated at the second target position **70**, the controller **600** may stop the printing process.

[0111] FIG. **17** is a schematic view illustrating an example of a first full image further including a third image of FIG. **15**. FIG. **18** is a schematic view illustrating an example of a second full image further including a fourth image of FIG. **16**.

[0112] Referring to FIGS. **2**, **15** and **17**, the first camera **300** may acquire a first full image **301** corresponding to the first area **51a** of the substrate **50** by photographing the substrate **50**.

[0113] Since the second target position **70** may be positioned in the first area **51a** of the substrate **50**, the first full image **301** acquired by the first camera **300** may further include the third image **301b**.

[0114] Referring to FIGS. **2**, **16** and **18**, the second camera **400** may acquire a second full image **401** corresponding to the first area **51a** of the substrate **50** by photographing the substrate **50**.

[0115] Since the second target position **70** may be positioned in the first area **51a** of the substrate **50**, the second full image **401** acquired by the second camera **400** may further include the second image **401b**.

[0116] FIGS. **19** to **22** are schematic views illustrating a manufacturing method for a display device according to an embodiment.

[0117] First, referring to FIG. **19**, an anode electrode **54** may be formed on a base substrate **52**. Although not illustrated in the views, a thin film transistor layer may be formed between the base substrate **52** and the anode electrode **54**. The thin film transistor layer may include at least one thin film transistor and an insulating layer.

[0118] A pixel defining layer **53** having an opening exposing the anode electrode **54** may be formed on the base substrate **52**.

[0119] Referring to FIGS. **20** and **21**, an emission layer **55** may be formed by discharging a first droplet **10** including an organic light emitting material into the opening of the pixel defining layer **53**. The first droplet **10** may be discharged by the droplet discharging apparatus **1000** of FIGS. **1** to **4**. The opening may mean the first target position (e.g., the first target position **60** of FIG. **1**).

[0120] Before the emission layer **55** may be formed, the base substrate **52** may be moved in the first direction **D1**. For example, the transfer part (e.g., the transfer part **100** of FIG. **2**) may move the base substrate **52** in the first direction **D1**. Accordingly, the first camera **300** and the second camera **400** may photograph the opening of the pixel defining layer **53** at different timings. For example, the first camera **300** may photograph the opening of the pixel defining layer **53** earlier than the second camera **400**.

[0121] Specifically, as illustrated in FIG. **20**, the first camera **300** may acquire a first image corresponding to the opening of the pixel defining layer **53** at a first timing. At the first timing, the base substrate **52** may be moving in the first direction **D1** by the transfer part **100**.

[0122] The first image may be an image corresponding to the opening of the pixel defining layer **53** before the first droplet **10** may be discharged. In other words, a scene in which the first droplet **10** does not impact the opening of the pixel defining layer **53** may appear in the first image.

[0123] As illustrated in FIG. **21**, the inkjet head **200** may discharge the first droplet **10** at a second timing later than the first timing. The first droplet **10** may impact the opening of the pixel defining layer **53**. Accordingly, the emission layer **55** may be formed in the opening of the pixel defining layer **53**. At the second timing, the base substrate **52** may be moving in the first direction **D1** by the transfer part **100**.

[0124] Referring to FIG. **22**, the second camera **400** may acquire a second image corresponding to the opening of the pixel defining layer **53** at a third timing later than the second timing. At the third

timing, the base substrate **52** may be moving in the first direction **D1** by the transfer part **100**.

[0125] The second image may be an image corresponding to the opening of the pixel defining layer **53** after the first droplet **10** may be discharged. In other words, a scene in which the first droplet **10** impacts the opening of the pixel defining layer **53** may appear in the second image.

[0126] The image processor **500** (e.g., the image processor **500** of FIG. **1**) may receive the first image and the second image, and compare the first image and the second image to inspect a discharge state (e.g., impact or not) of the first droplet **10**. For example, the image processor **500** may compare the first image and the second image to check whether the first droplet **10** may be properly seated at the first target position **60**. In case that it is determined that the first droplet **10** is not seated at the opening of the pixel defining layer **53**, the controller **600** (e.g., the controller **600** of FIG. **1**) may stop the printing process.

[0127] In the related art, a state of the droplet was inspected before performing the printing process, but there is a problem in that it is not possible to check whether the droplet is impacted during the printing process.

[0128] However, according to the droplet discharging apparatus **1000**, the droplet discharging method, and the manufacturing method of the display device according to embodiments of the disclosure, in the time period during which the printing process may be performed, an image (e.g., the first image **301a**) corresponding to a target position (e.g., a first target position **60**) may be acquired immediately before a droplet (e.g., a first droplet **10**) may be discharged to the target position, another image (e.g., the second image **401a**) corresponding to the target position may be acquired immediately after the droplet may be discharged to the target position, and the images (e.g., the first image **301a** and the second image **401a**) may be compared. Accordingly, it may be possible to check in real time whether the droplet may be impacted while the printing process may be performed.

[0129] If it is determined that the droplet is not seated at the target position, the printing process may be stopped. Accordingly, it may be possible to minimize the product to be thrown away due to poor impact, and unnecessary processes may not be performed.

[0130] Although certain embodiments and implementations have been described herein, other embodiments and modifications will be apparent from this description. Accordingly, the disclosure is not limited to such embodiments, but rather to the broader scope of the appended claims and various obvious modifications and equivalent arrangements as would be apparent to a person of ordinary skill in the art.

Claims

1. A manufacturing method for a display device, the method comprising: forming an anode electrode on a base substrate; forming a pixel defining layer on the base substrate, the pixel defining layer having an opening exposing the anode electrode; moving the base substrate in a first direction; acquiring a first image corresponding to the opening of the base substrate; forming an emission layer by discharging a first droplet including an organic light emitting material into the opening of the base substrate; acquiring a second image corresponding to the opening of the base substrate; and comparing the first image and the second image to inspect a discharge state of the first droplet.
2. The method of claim 1, wherein the first image is an image corresponding to the opening before the first droplet is discharged, and the second image is an image corresponding to the opening after the first droplet is discharged.
3. The method of claim 1, wherein the first image is acquired at a first timing, the first droplet is discharged at a second timing, and the second image is acquired at a third timing.
4. The method of claim 3, wherein at the first timing, the second timing, and the third timing, the base substrate is being moved in the first direction.

5. The method of claim 1, further comprising: stopping a printing process in case that the first droplet is not seated at the opening.

6. The method of claim 1, further comprising: acquiring a third image corresponding to a second target position of the opening of the base substrate, the second target position spaced apart from a first target position in a second direction opposite to the first direction, the first target position corresponding to the first image and the second image; discharging a second droplet to the second target position of the opening of the base substrate; acquiring a fourth image corresponding to the second target position of the opening of the base substrate; and comparing the third image and the fourth image to inspect a discharge state of the second droplet.
